

Electrical Instrumentation Part-II

PowerPack Phase 5

Waveform Display Devices and Applications

CRT, LED, LCD and Display Device Comparison

Diagram-safe, definition-first, semester-answer style

Semester Command

This Phase 5 PowerPack is for the display-device part of Electrical Instrumentation. Learn every answer in this order:

Definition → Construction → Working Principle → Diagram → Applications → Advantages and Limitations

For every diagram, keep wires straight and perpendicular. Labels must be far from wires and components. If a diagram becomes large, split it into smaller diagrams.

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1. Phase 5 Target: What You Must Be Able To Answer

After completing this phase, you must be able to answer:

1. What is a waveform display device?
2. Explain CRT construction and working.
3. Explain why CRT is used in CRO.
4. Explain fluorescent screen, electron gun and deflection plates.
5. Explain LED construction and working.
6. Explain seven-segment LED display.
7. Explain LCD construction and working.
8. Compare CRT, LED and LCD.
9. Write applications of CRT, LED and LCD.
10. Answer short notes on display devices in semester style.

Must Know

Display-device questions are usually short but diagram-based. A clean labelled diagram can carry a large part of the marks.

2. Waveform Display Devices from Absolute Zero

A waveform display device is an output device which visually represents electrical information.

In instrumentation, the measured quantity is first converted into an electrical signal. This signal must be displayed so that the observer can read or analyse it.

Measured quantity → electrical signal → visual display

Examples of display devices are:

CRT, LED display, LCD display

2.1. Need of Display Devices

Display devices are required for:

1. observing waveforms,
2. measuring voltage and time,
3. displaying numerical readings,
4. showing instrument status,
5. displaying digital output from ADC,
6. monitoring industrial and laboratory systems.

2.2. Basic Display System

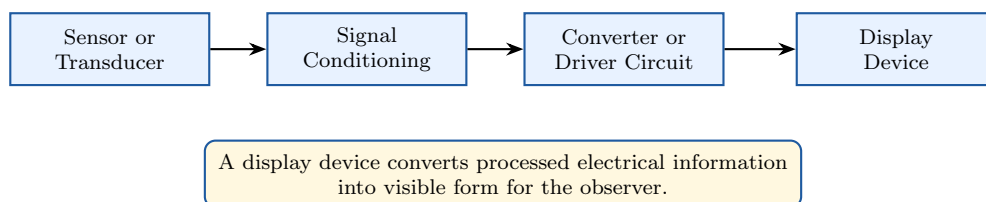


Figure 1: Basic display system in electrical instrumentation.

3. Cathode Ray Tube

3.1. Definition

A Cathode Ray Tube, or CRT, is a vacuum tube in which an electron beam is produced, accelerated, focused, deflected and allowed to strike a fluorescent screen to produce a visible spot.

CRT converts electrical deflection signals into visible motion of a bright spot.

In a CRO, the CRT is the main display device.

3.2. Main Parts of CRT

A CRT consists of the following parts:

1. Electron gun,
2. Deflection plate system,
3. Fluorescent screen,
4. Evacuated glass envelope,
5. Aquadag coating,
6. Base pins and external connections.

3.3. Clean CRT Diagram

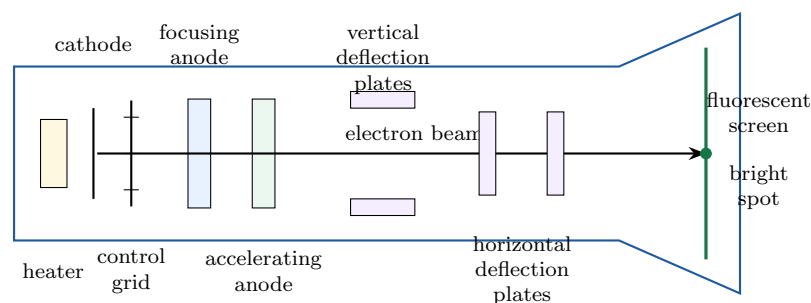


Figure 2: Constructional diagram of a cathode ray tube.

4. Working of CRT

The working of CRT can be understood block by block.

4.1. Step 1: Electron Emission

The heater heats the cathode. Due to thermionic emission, electrons are emitted from the cathode.

Heated cathode emits electrons.

4.2. Step 2: Intensity Control

The control grid controls the number of electrons reaching the screen. If the grid is made more negative, fewer electrons pass through it.

Therefore, the brightness of the spot decreases.

Control grid controls brightness.

4.3. Step 3: Focusing

The focusing anode focuses the electron beam into a fine beam. This produces a sharp spot on the screen.

Focusing anode controls sharpness.

4.4. Step 4: Acceleration

The accelerating anode gives high velocity to electrons so that they strike the fluorescent screen with sufficient energy.

Accelerating anode controls electron speed.

4.5. Step 5: Deflection

The vertical plates deflect the electron beam up or down. The horizontal plates deflect the electron beam left or right.

Vertical plates → voltage axis

Horizontal plates → time axis

4.6. Step 6: Fluorescent Screen

When high-speed electrons strike the fluorescent screen, light is emitted. This produces a visible spot.

Electron beam + screen $\rightarrow$ visible spot

5. Electron Gun

5.1. Definition

The electron gun is the part of CRT which produces, controls, focuses and accelerates the electron beam.

It consists of:

heater, cathode, control grid, focusing anode and accelerating anode

5.2. Electron Gun Diagram

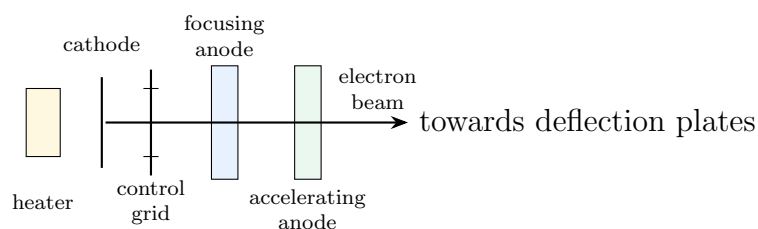


Figure 3: Electron gun of a CRT.

5.3. Functions

Part	Function
Heater	Heats the cathode.
Cathode	Emits electrons by thermionic emission.
Control grid	Controls beam intensity and brightness.
Focusing anode	Focuses the electron beam into a fine spot.
Accelerating anode	Accelerates electrons towards the screen.

6. Electrostatic Deflection in CRT

6.1. Basic Principle

In a CRT, electrostatic deflection is produced by applying voltage between a pair of deflection plates.

If voltage is applied to vertical plates, the electric field deflects the electron beam vertically.

If voltage is applied to horizontal plates, the electric field deflects the electron beam horizontally.

Deflection is proportional to applied plate voltage.

6.2. Deflection Plate Diagram

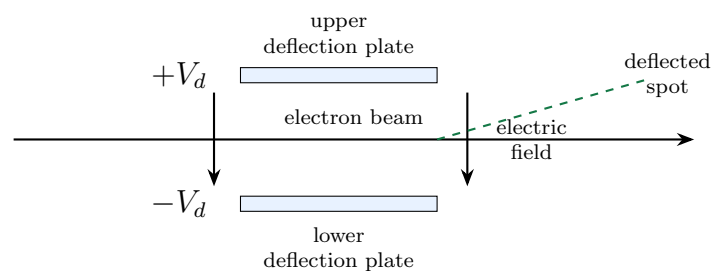


Figure 4: Electrostatic deflection of electron beam in CRT.

6.3. Deflection Sensitivity

Deflection sensitivity is defined as deflection on the screen per unit deflecting voltage.

$$S_d = \frac{\text{deflection on screen}}{\text{deflecting voltage}}$$

Unit:

$$\text{m/V or mm/V}$$

A high deflection sensitivity means a small voltage produces a large movement of the spot.

7. Fluorescent Screen and Persistence

7.1. Fluorescent Screen

The inner surface of the CRT screen is coated with fluorescent material. When electrons strike this material, light is emitted.

Electron energy is converted into light energy.

7.2. Persistence

Persistence is the time for which light continues to be emitted from the screen after the electron beam is removed.

Persistence = afterglow duration of screen phosphor

7.3. Importance of Persistence

1. It helps in observing waveforms clearly.
2. Too low persistence gives faint display.
3. Too high persistence causes smearing of fast waveforms.
4. Medium persistence is suitable for general CRO use.

7.4. Screen Behaviour Diagram

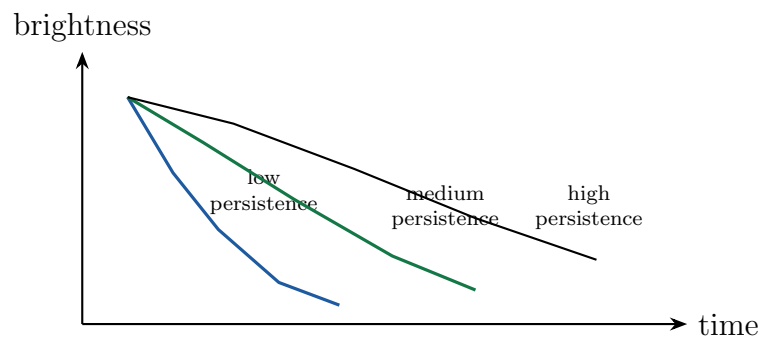


Figure 5: Effect of phosphor persistence on screen brightness.

8. LED Display

8.1. Definition

An LED, or Light Emitting Diode, is a semiconductor diode which emits light when it is forward biased.

LED converts electrical energy into light energy.

8.2. Working Principle

When an LED is forward biased, electrons and holes recombine at the p-n junction. During recombination, energy is released in the form of light.

electron-hole recombination $\rightarrow$ light emission

The colour of emitted light depends on the semiconductor material and energy band gap.

8.3. LED Symbol and Connection

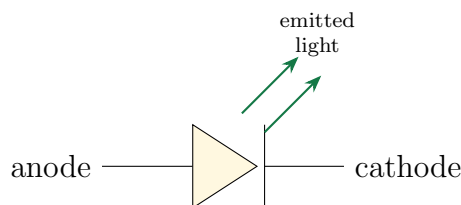


Figure 6: LED symbol showing light emission.

8.4. Advantages of LED Displays

1. Low operating voltage.
2. Fast response.
3. Long life.
4. Small size.
5. High reliability.
6. Good visibility in dark environments.

8.5. Limitations of LED Displays

1. Consumes more power than LCD.
2. Brightness depends on current.
3. Requires current limiting resistor.
4. Not ideal for very low-power battery instruments.

9. Seven-Segment LED Display

9.1. Definition

A seven-segment display is a display device made of seven LED segments arranged in the form of digit 8. By switching selected segments ON and OFF, decimal digits from 0 to 9 can be displayed.

The seven segments are usually named:

a, b, c, d, e, f, g

9.2. Seven-Segment Diagram

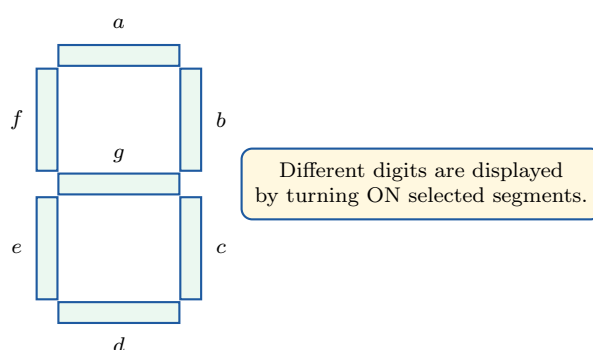


Figure 7: Seven-segment LED display.

9.3. Common Anode and Common Cathode

Type	Meaning
Common anode display	All anodes are connected together. A segment glows when its cathode is made LOW.
Common cathode display	All cathodes are connected together. A segment glows when its anode is made HIGH.

9.4. Digit Formation Table

Digit	Segments ON
0	a, b, c, d, e, f
1	b, c
2	a, b, g, e, d
3	a, b, c, d, g
4	f, g, b, c
5	a, f, g, c, d
6	a, f, e, d, c, g
7	a, b, c
8	a, b, c, d, e, f, g
9	a, b, c, d, f, g

10. LCD Display

10.1. Definition

LCD stands for Liquid Crystal Display. It is a display device that uses liquid crystal material to control the passage of light.

LCD does not produce light by itself; it controls transmitted or reflected light.

10.2. Basic Principle

Liquid crystals are materials having properties between solids and liquids. Their optical behaviour changes when an electric field is applied.

When voltage is applied across the liquid crystal layer, the orientation of molecules changes. This changes the polarisation of light and controls whether light passes through the display.

electric field changes crystal orientation $\rightarrow$ light transmission changes

LCD Construction Diagram

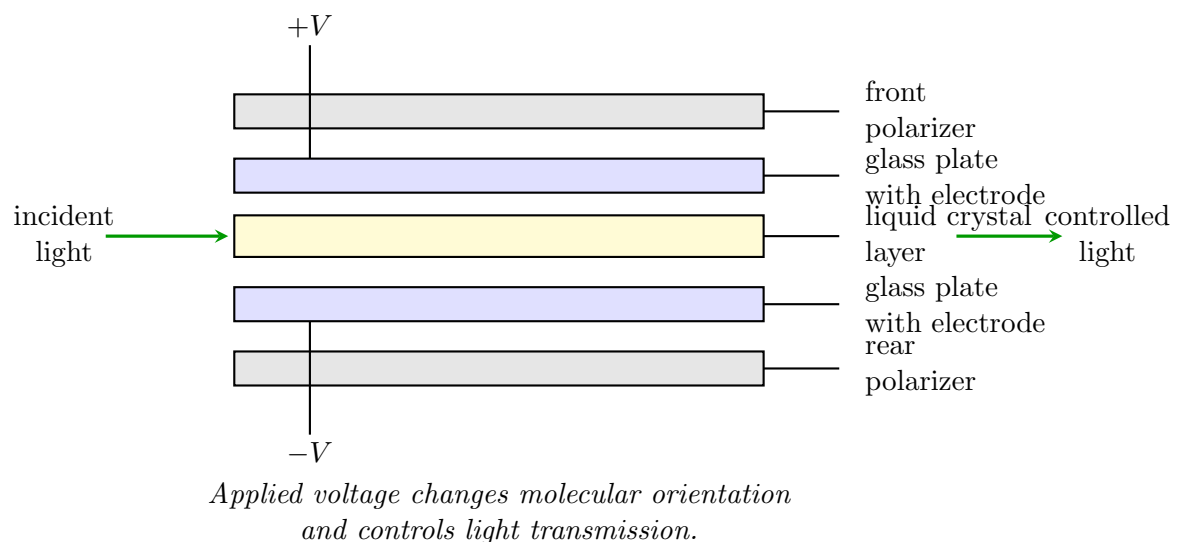


Figure 8: Basic construction of a liquid crystal display.

11. Working of LCD

11.1. Without Applied Voltage

In the absence of applied voltage, liquid crystal molecules have a natural twisted arrangement. This arrangement rotates the plane of polarisation of light.

Depending on polarizer orientation, light may pass through the display and the segment appears bright.

No voltage → natural molecular orientation

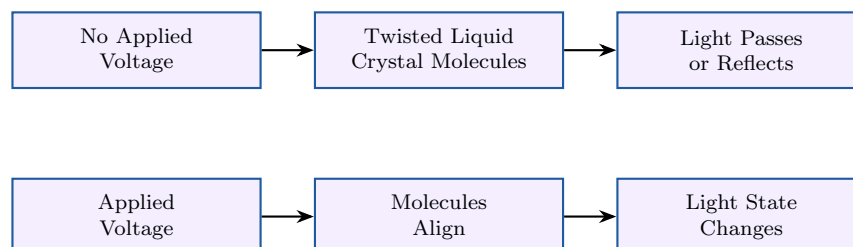
11.2. With Applied Voltage

When voltage is applied, liquid crystal molecules align with the electric field. Their ability to rotate polarised light changes.

As a result, light transmission changes and the selected segment appears dark or bright depending on the display type.

Applied voltage → changed molecular orientation

11.3. LCD Segment Working Diagram



LCD controls light; it is not a self-luminous display.

Figure 9: Working principle of LCD.

11.4. Advantages of LCD

1. Very low power consumption.
2. Thin and light in weight.
3. Suitable for battery-operated instruments.
4. Low operating voltage.

5. Long life.
6. Good for digital display panels.

11.5. Limitations of LCD

1. Slower response than LED.
2. Poor visibility in darkness without backlight.
3. Limited viewing angle.
4. Temperature affects performance.
5. Cannot produce light by itself.

12. Comparison of CRT, LED and LCD

Point	CRT	LED	LCD
Full form	Cathode Ray Tube	Light Emitting Diode	Liquid Crystal Display
Working principle	Electron beam strikes fluorescent screen	Electron-hole recombination emits light	Electric field controls light transmission
Self luminous	Yes	Yes	No
Power consumption	High	Moderate	Very low
Size	Bulky	Small	Very thin
Response speed	High	Very high	Moderate or slow
Used in	CRO, old monitors	Indicators, seven-segment display	Calculators, digital meters, panels
Visibility in dark	Good	Good	Needs backlight

12.1. Most Important Comparison Lines

CRT uses electron beam.

LED emits light due to recombination.

LCD controls light but does not generate light.

Semester Command

For comparison questions, do not write only paragraphs. Use a table. A comparison table is the fastest way to secure marks.

13. Applications of Display Devices

13.1. Applications of CRT

1. Cathode Ray Oscilloscope.
2. Radar display.
3. Old television receivers.
4. Old computer monitors.
5. Waveform analysis instruments.

13.2. Applications of LED

1. Indicator lamps.
2. Seven-segment displays.
3. Digital clocks.
4. Instrument panels.
5. Alarm and status displays.
6. Optical communication sources.

13.3. Applications of LCD

1. Digital multimeters.
2. Calculators.
3. Watches.
4. Mobile displays.
5. Instrument display panels.
6. Low-power portable devices.

14. Short Notes in Semester Style

14.1. Short Note: CRT

A Cathode Ray Tube is a vacuum display device in which a focused electron beam is generated by an electron gun and made to strike a fluorescent screen. The electron beam is deflected by electrostatic deflection plates. The vertical deflection corresponds to applied signal voltage and horizontal deflection corresponds to time in a CRO. Hence CRT is very useful for waveform display.

CRT is the heart of CRO.

14.2. Short Note: LED Display

An LED display uses light emitting diodes as display elements. When the LED is forward biased, electrons and holes recombine at the junction and energy is released as light. LED displays are bright, fast, reliable and suitable for indicators and seven-segment displays. However, they consume more power than LCD.

LED is a self-luminous semiconductor display.

14.3. Short Note: LCD

An LCD is a low-power display device using liquid crystal material. The orientation of liquid crystal molecules changes when an electric field is applied, thereby controlling light transmission. LCD does not produce light by itself. It is widely used in digital meters, calculators and portable instruments because of low power consumption.

LCD controls light; it does not generate light.

15. How to Recognise Display Device Questions

Question Language	What You Must Write
“Explain CRT with diagram”	Draw CRT construction, then explain electron gun, deflection plates and screen.
“Functional parts of CRT”	Write heater, cathode, grid, focusing anode, accelerating anode, plates, screen.
“Why CRT is used in CRO”	Write electron beam deflection, waveform display, voltage-time representation.
“Explain LED display”	Define LED, draw symbol, explain forward bias and recombination.
“Seven-segment display”	Draw seven segments <i>a</i> to <i>g</i> , write common anode/cathode and segment table.
“Explain LCD”	Draw layered construction, explain liquid crystal molecule orientation and light control.
“Compare CRT, LED and LCD”	Write table: working principle, power, size, speed, application.

16. One-Page Formula and Definition Bank

CRT = electron beam display device

Electron gun = source, controller, focuser and accelerator of electron beam

$$\text{Deflection sensitivity} = \frac{\text{screen deflection}}{\text{deflecting voltage}}$$

Persistence = duration of afterglow of phosphor screen

LED = forward-biased p-n junction emitting light

LCD = liquid crystal device controlling light transmission

Seven-segment display = seven LED segments arranged to display digits

LCD consumes less power than LED and CRT.

CRT is bulky but excellent for waveform display.

17. Final Rapid Revision for Phase 5

17.1. If Only 20 Minutes Are Left

Revise in this order:

1. CRT construction diagram.
2. Functions of electron gun parts.
3. Electrostatic deflection principle.
4. LED definition, symbol and working.
5. Seven-segment display diagram.
6. LCD construction and working.
7. CRT vs LED vs LCD comparison table.

17.2. Most Probable Diagram Commands

1. CRT constructional diagram.
2. Electron gun diagram.
3. Electrostatic deflection plate diagram.
4. LED symbol.
5. Seven-segment display.
6. LCD layered construction.

Do Not Write This Mistake

Do not draw tiny, crowded diagrams. In display-device answers, clean labelling is more important than artistic decoration. Put labels outside the main diagram and connect only when needed.

Phase 5 Completed: Display Device Mastery